

SANKUR KUNDU

Software Engineer | Backend, Systems & Full-Stack

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SUMMARY

Software engineer (B.Tech CSE, 2026) who builds systems from scratch and proves them correct: **relay** (a fault-tolerant LLM inference cluster in **C++/CUDA** and **Swift**), **ratchet** (RL post-training: **GSM8K 48% → 52.5%**), **quorumdb** (TLA+-verified SQL), **tessel** (GPU kernels at **cuBLAS** parity). Co-founded Gigzen as its sole engineer.

PROJECTS

relay — C++, CUDA and Swift: a disaggregated, fault-tolerant LLM inference cluster

- Built **C++/CUDA** prefill/decode workers with my own **flash-attention** and **flash-decoding** kernels, a **KV transfer engine** streaming each layer as the next computes, and a **Swift** control plane (**C++ interop**, OpenAI API, **prefix-aware routing**).
- On a **T4**, my kernels cut 2,048-token prefill **8.7×** (**3.7 s → 0.43 s**) and batch-32 decode **5.4×**; streaming hides **99–100%** of the KV transfer; output matches **Hugging Face**, **bit-identical** whether batched, cached, preempted or split across workers.
- Fault-tolerant**: killed, frozen or stalled workers are caught by heartbeat and progress watchdogs; their requests **resume elsewhere with identical output** (**384/384** through **60 SIGKILLs**; no request lost in **300** fault-injection seeds).

ratchet — Python, PyTorch and C++/CUDA: reinforcement-learning post-training (GRPO) from scratch

- Built **GRPO** post-training end to end: rollouts with per-token log-probabilities on **relay** (my C++/CUDA engine, through a C API), a **PyTorch** trainer data-parallel on **tandem**, **in-place GPU weight sync**, **partial rollouts** and **one-step-ahead** training.
- On **2× T4**: **Qwen2.5-0.5B** on **GSM8K 48.1% → 52.5%** in 100 steps; weight sync **369×** faster than a reload; found and fixed **fp16** rounding that silently erased training updates (log-prob gap **5.2 → 0.04**).

quorumdb — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible

- Built from scratch a **distributed SQL database** that **psql** connects to: **LSM storage**, **Raft**, a self-rebalancing **multi-Raft** store, **serializable ACID transactions** (Percolator), a SQL planner; **1,500 tpmC** on a **TPC-C**-derived benchmark.
- Proved by **deterministic simulation** (**10,000 Raft clusters**, **610K crashes**, histories machine-checked) and a **TLA+** model (**337K states**); found **6 real bugs**, catches all **15 planted**; **time-travel debugger** rewinds runs by replay.

Also built: tandem • tessel • faultline • kiln • ferrolm

- tandem**, distributed training (DDP, ZeRO-1/2/3, 1F1B) **bit-identical to PyTorch** on **2× T4**; **tessel**, a tile language for GPUs and TPUs with GEMM at **parity with cuBLAS**; **kiln**, an ML compiler **faster than torch.compile** and **JAX/XLA** on CPU.
- faultline**, Jepsen-style fault injection that tested **4 etcd release lines** (**1.26M** operations under **288** faults); **ferrolm**, an LLM server with a paged KV cache and continuous batching (**3.4×** throughput).

OPEN SOURCE

- PyTorch #199875** (under review): **torch.compile** now fills **torch.empty** in deterministic mode like eager, without filling **Inductor**'s internal buffers, the cause of an earlier fix's revert.
- PyTorch #199441**, **#199646** (under review): removed **13 stale torch.compile test skips** (verified on CPU and **2× T4**); **8** skipped determinism tests now run under **torch.compile**.

EXPERIENCE

Gigzen — Co-founder & Founding Engineer

Apr 2026 – Present

Bhubaneswar, India • Early-stage startup (incorporation in progress) • Sole engineer across every product

- Sole engineer** on **Waggle**, a delivery platform of **three Android apps** (riders, shops, customers) from one **TypeScript / React / Capacitor** codebase on **Supabase PostgreSQL**: **53 tables**, **90 RLS policies**, **154 SQL functions**.
- Put the business rules in the database, where no client can bypass them: **first-accept-wins dispatch** (45-second offer window), **4-digit handover codes**, **Razorpay/UPI** payments reconciled by UTR, **GST invoices** and **pg_cron** billing runs.
- Security-tested** it with **150 automated attacks** on its database access rules (all blocked) and a **42-step Playwright** run.
- Built **Populace**, an open-source multi-user testing tool (**Node.js**, an **npm CLI**, **GitHub Action** and **Electron** app); its **first run** found **5 defects in 3.5 minutes** after manual testing had passed, one breaking **every signup**.

SKILLS

CS Fundamentals: Data Structures & Algorithms, System Design, Distributed Systems, DBMS internals, OS, Networks

Languages: C++, Rust, Python, Swift (C++ interop), TypeScript, JavaScript, SQL (PL/pgSQL), Java

Backend & Data: PostgreSQL, Supabase, Row-Level Security, pg_cron, REST APIs, Node.js, FastAPI, Flask, Redis, Razorpay/UPI

Frontend & Mobile: React, Capacitor (Android), Electron, Leaflet/MapLibre maps, HTML, CSS

Systems & performance: CUDA (tensor cores), LLM serving (disaggregated prefill/decode), RL post-training, DDP/ZeRO, SIMD

Testing, DevOps & Cloud: Simulation testing, fault injection, TLA+, Playwright, GitHub Actions, Docker, Kubernetes, GCP

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata

Sep 2022 – Apr 2026